6 In this question you must show detailed reasoning.

You are given that $f(x) = 4x^3 - 3x + 1$.

(i) Use the factor theorem to show that $(x+1)$ is a factor of $f(x)$. **[2]**

(ii) Solve the equation $f(x) = 0$. **[3]**

6 (i)	
6 (ii)	